

Zardar Khan

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SUMMARY

I build ML systems in healthcare—messy data, tight timelines, real consequences. 4+ years shipping production pipelines: constrained decoding (180x speedup), clinical RAG (3 weeks to trial), PHI de-identification (45% → 90% F1). I reverse-engineer undocumented systems and make them work.

WORK

- Constrained Decoding for Medical JSON** | [code](#) 2025
- Pathology reports → structured JSON. Prompt-only approaches hit ~67% validity. Not good enough at scale.
 - Built two-stage pipeline: reason first, then constrained emit. Monkey-patched xgrammar into llama.cpp (27x), then vLLM migration with continuous batching (7x more).
 - 180x speedup (3 min → 1 sec). 100% schema validity. 17K reports in 5 hrs. Scaling to 1M+.
- Caring Contacts RAG** 2024
- Discharge letters for psychiatric patients that reference specific details from their stay.
 - BM25 → reranker. Proposition-level chunking beat doc/paragraph for retrieval. Prompt guardrails for safety.
 - Concept to clinical trial in 3 weeks. Presented ISBD 2025 (Japan), IASR 2025 (Boston).
- Mammography Classification** | [code](#) 2024
- Sort DICOMs by artifact type (mag view, biopsy tool). Metadata too inconsistent to trust.
 - EfficientNet + active learning: 50 labels → 90% accuracy on 400 samples.
- PHI De-identification** 2024
- MIMIC/i2b2 models miss hospital-specific patterns—too generic for our data.
 - Built synthetic data pipeline: SQL → sample real PHI patterns → GPT-4 generates text with XML placeholders → swap identifiers back in → train NER.
 - 45% → 75% F1. Mentored student who pushed to 90%.
- FHIR Extraction** 2023
- Needed 2M+ clinical reports. Existing queries took 120 days.
 - Reverse-engineered undocumented Observation → DiagnosticReport structure. Temporary Postgres indexes.
 - 120 days → 4 hours (700x). Found 500K missing reports from import errors.

JOBS

- Sunnybrook Research Institute** Sept. 2023 – Present
Research Software Engineer Toronto, ON
- Sole ML engineer across 3 research labs. Built all pipelines above. Go-to for pathology/radiology report extraction—wrote custom parsing to stitch fragmented reports that appear line-by-line.
 - Routinely pull and create datasets with thousands of reports for downstream research. Work adopted by 2 external labs.
 - Quantized RadBERT classifier: 450ms → <100ms (4.5x), 95% accuracy. Oral presentation at IGTxIMNO 2024.
- gojitech** Aug. 2021 – Sept. 2023
Software Engineer Toronto, ON
- Real-time transcription (WebSocket + Azure STT, replaced 30s batch). Medication extraction (GPT-Neo).
 - Fixed Chrome 2GB memory bug killing long recordings.

SIDE PROJECTS

- Bookkeeping Agent** — Built tool-use agent from scratch. Hit context limits on large files, calculator failed on OCR errors, pivoted to DuckDB for SQL on CSVs. Found I was spending \$70/mo on Starbucks.
- Semantic Routers for MoE** — Cohere fellowship pitch. Proposed learnable expert embeddings over linear routers. Got rejected. Revisiting at tiny scale.
- Pronunciation App** — LLM-as-judge for my daughter's reading. Customer base of 1, retention 100%.

EDUCATION

Toronto Metropolitan University

Toronto, ON

Bachelor of Engineering, Biomedical Engineering

PUBLICATIONS

Khan Z et al. *AI-Personalized Caring Contacts*. IASR 2025 (Boston), ISBD 2025 (Japan).

Khan Z et al. *RadBERT Microcalcification Classifier*. IGTxIMNO 2024 — Oral Presentation.

SKILLS

LLMs & NLP: Transformers · llama.cpp · vLLM · RAG · Tool-use Agents · Constrained Decoding · NER

ML Techniques: Quantization · Active Learning · Knowledge Distillation · Synthetic Data Generation

Stack: PyTorch · FHIR · Postgres · Docker · AWS/Azure · FastAPI